

IDX EXCHANGE

Case Study Project - Data Analyst Internship

Did Compass Win California by Recruiting Top Agents?

A market-share and talent-migration investigation using CRMLS Sold transaction data through 2025

At a Glance

Format: Independent case study, individual or pairs

Duration: 4 weeks (suggested pacing below)

Tools: SQL, Python (pandas), and a charting library of your choice

Dataset: CRMLS Sold transactions via the CoreLogic Trestle API pipeline, Residential SFR, through 2025 — historical start year is limited by CRMLS access (see Data Access)

Final outputs: 6 deliverables + Executive Memo + 10-minute readout

Confidentiality: Real, live MLS transaction data — internal program use only, do not share externally

Overview

Between 2018 and 2023, Compass grew from a minor player into one of the largest residential brokerages in California by transaction volume. The open question this case study asks you to investigate, with data rather than narrative, is how that growth actually happened: by recruiting already-successful agents away from legacy brokerages, by acquiring smaller firms outright, by growing organically through new and lower-producing agents, or some combination of the three.

This matters beyond the history lesson. IDX Exchange works with brokerages and agent teams who are constantly evaluating recruiting, retention, and growth strategy. The analytical pattern you'll build here — agent-level migration tracking, brokerage market share over time, and acquisition detection from disappearing entities — is the same pattern used in real client-facing competitive intelligence work.

Learning Objectives

- Design and execute a multi-year panel analysis from raw transaction-level data.
- Build entity-resolution logic to track individual agents and offices across a changing competitive landscape.
- Apply a difference-in-differences framework to isolate the effect of switching brokerages from other confounding factors.
- Communicate a data-driven argument to a non-technical audience through an executive memo and live readout.

Background & Business Context

Real estate brokerages compete on three levers: technology, commission splits, and brand — but ultimately, transaction volume follows agents, not logos. When a high-producing agent switches brokerages, their listings, their client relationships, and often their team move with them. This makes agent migration one of the highest-signal indicators of competitive momentum in residential real estate, and one of the hardest to see without transaction-level data.

Compass is a useful case because its growth was fast and visible enough that the underlying mechanism is debated publicly: some attribute it to technology and marketing spend, others to an aggressive, well-capitalized recruiting strategy targeting top producers at Keller Williams, Coldwell Banker, and RE/MAX. Your job is to settle that debate with data, for California specifically.

Dataset & Technical Setup

You will work from CRMLS Sold transactions through 2025, filtered to Residential Single Family Residence, going as far back as CRMLS access allows — confirm the earliest available year in Week 1 (see Data Access). The same field structure underlies IDX Exchange's production property databases, so the conventions you build here (composite-key joins over raw name joins, explicit handling of brokerage rebrands) are the same conventions used internally.

Required Fields

Field	Type	Notes
CloseDate / ListingContractDate / PurchaseContractDate	Date	Trestle export field names. Use CloseDate to bucket transactions by year. Confirm exact formats during your Week 0–style metadata review before writing joins.
ListOfficeName / BuyerOfficeName	String	Brokerage/office display name — this is the field this case study's market-share and migration logic runs on. Normalize variants (e.g., “Compass” vs “Compass RE”) before aggregating.
ListAgentFullName / BuyerAgentFullName	String	Trestle/CRMLS exports typically expose agent name only, not a stable agent ID. See “Agent & Office ID Resolution” below — this affects every agent-level deliverable.
ClosePrice / ListPrice / OriginalListPrice	Numeric (USD)	ClosePrice drives listing volume and luxury-concentration cuts (Deliverables 1, 3, 6); OriginalListPrice supports negotiation-strength metrics if you extend the analysis.
CountyOrParish / MLSAreaMajor	String	Geographic grouping fields, matching the naming convention used elsewhere in the IDX Exchange MLS pipeline.
PropertyType / PropertySubType	String	Filter to PropertyType == ‘Residential’ and PropertySubType == Single Family Residence, per project scope.
Latitude / Longitude	Numeric	Optional, for geographic QA or office-location breakouts. Flag null, zero (sentinel), and positive-longitude (sign error) values — see Data Quality Checks below.

Agent & Office ID Resolution — Read First

Trestle/CRMLS exports generally expose ListAgentFullName and ListOfficeName, not a stable agent or office ID. Before building any migration matrix, top-producer ranking, or acquisition-detection logic, build a normalized composite key (e.g., lowercased, trimmed full name + office name) as an explicit entity-resolution step, and document every normalization rule you apply. This single decision determines whether Deliverables 2, 3, 5, and 6 are valid.

Data Access

This case study runs on the same CRMLS pipeline that feeds IDX Exchange's production data: a CoreLogic Trestle API extraction process that exports monthly CRMLSSoldYYYYMM.csv files of closed transactions. This case study uses the Sold data only — closed transactions give you both volume (\$) and units (count of closed sides), which is everything the market-share, migration, and concentration deliverables need. You do not need the active Listing files for this project.

- Pre-generated files: pulled via the same FTP server used in Week 0 of the Data Analyst track onboarding. Request current login credentials from your track lead — don't reuse old ones, as they rotate periodically.

- **Known coverage gap:** the standard pipeline only pre-generates files from January 2024 forward. This case study ideally wants 2015–2025. For 2015–2023, you will need to either (a) re-run the Trestle extraction script with an earlier date range, per the Week 0 orientation materials, or (b) request a historical CRMLS archive export from the data team.
- **CRMLS access limit:** CRMLS restricts how far back historical data can be pulled, and that limit may fall short of 2015 regardless of which extraction path you use. Don't assume the ten-year window is achievable. In Week 1, run the extraction (or archive request) first and confirm the earliest year actually returned — then treat that year, not 2015, as the true start of your panel, and state it explicitly in your methodology notes and Executive Memo. A shorter, accurate window is far better than a fabricated or silently truncated one.
- Filter to `PropertyType == 'Residential'` immediately after load, before any other processing, consistent with the rest of the IDX Exchange MLS pipeline.

Data Quality Checks Before You Start

- Confirm `ListOfficeName` / `BuyerOfficeName` spellings are stable over time; some MLS feeds introduce new variants after a rebrand or merger — reconcile these manually before building migration matrices.
- Deduplicate listing sides where a single closed transaction has co-listing agents, to avoid double-counting volume.
- Spot-check a known Compass office acquisition (e.g., a regional luxury boutique) to validate that your office-resolution logic captures it correctly.
- Build date-consistency flag columns before any time-series work — `listing_after_close_flag`, `purchase_after_close_flag`, and `negative_timeline_flag` — and report (don't silently drop) any records that fail these checks.
- Run a missing-value pass on every required field; flag any column above 90% null and decide explicitly whether to drop or retain it. Retain core fields (`ClosePrice`, `CloseDate`, `ListOfficeName`) even if partially missing rather than dropping them outright.
- If you incorporate geographic breakouts, flag null or zero-value coordinates (sentinel nulls) and any positive-longitude records — California longitudes should be negative — before mapping anything.

Research Questions

1. When did Compass begin experiencing significant growth in California, and was the inflection point gradual or sudden?
 2. Which brokerages lost the most market share during that period, and were the losses concentrated or spread evenly across competitors?
 3. Were high-volume agents disproportionately moving to Compass relative to lower-producing agents?
 4. Did Compass's growth occur primarily through agent migration, brokerage acquisitions, organic growth of new agents, or a combination — and what is the approximate split?
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Deliverables

Each deliverable below includes the original scope plus a suggested method and expected output. You are not required to follow the suggested method exactly — if you have a better approach, use it and note why in your documentation.

Deliverable 1 — Brokerage Market Share Analysis

Calculate annual listing sides, listing volume (\$), and market share for Compass, Keller Williams, Coldwell Banker, RE/MAX, Berkshire Hathaway, Century 21, Sotheby's, eXp, Redfin, and an "Others" bucket. Produce a time-series chart of market share by listing volume, from your earliest available year through 2025.

Suggested Method

- Aggregate by ListOfficeName normalized to a fixed brokerage taxonomy (handle naming variants before grouping).
- Compute share = brokerage listing volume ÷ total California SFR listing volume, by year.
- Use a stacked area or multi-line chart; annotate the year Compass crosses each major competitor.

Deliverable 2 — Top Agent Migration Study

Identify the top 500 agents annually by listing volume and listing sides. Track brokerage affiliation changes year-over-year and build migration matrices showing which brokerages supplied agents to Compass.

Suggested Method

- Build an agent-year panel keyed on your normalized agent composite key, with a single "primary brokerage for the year" field (mode of ListOfficeName by transaction count).
- For each consecutive year pair, build an N×N transition matrix of brokerage → brokerage moves, restricted to the top-500 cohort.
- Surface the matrix as a heatmap, with a callout table for the top 5 source brokerages feeding Compass each year.

Deliverable 3 — Top Producer Concentration

Determine what percentage of the top 1%, 5%, and 10% of listing agents (by volume) were affiliated with Compass each year, from your earliest available year through 2025.

Suggested Method

- Rank all active agents annually by listing volume; compute percentile cutoffs.
- Calculate Compass's share of each percentile band and plot as three lines over time — if the top 1% line moves earlier and faster than the top 10% line, that is direct evidence of a recruiting-led (vs. organic) growth strategy.

Deliverable 4 — Office Growth Analysis

Analyze active office count, listing volume per office, and listings per office for Compass and major competitors.

Suggested Method

- Count distinct active offices (by normalized ListOfficeName) per brokerage per year, and divide total volume by office count to get a per-office productivity metric.
- Compare Compass's per-office productivity trend to competitors' — rising productivity per office suggests agent-quality-driven growth rather than simple office expansion.

Deliverable 5 — Acquisition Detection

Identify likely acquisitions by detecting brokerages that disappear from the data while their agents and production migrate to Compass within a short window.

Suggested Method

- Flag any normalized office (by ListOfficeName) whose transaction volume drops to zero within a calendar year.
- For each flagged office, check whether $\geq 50\%$ of its prior-year top agents (by volume) appear under a Compass ListOfficeName within the following 12 months — treat this as a likely acquisition rather than independent attrition.
- Cross-reference flagged events against public news where feasible, and clearly label which findings are data-inferred versus externally confirmed.

Deliverable 6 — Difference-in-Differences Style Analysis

Compare agents who moved to Compass versus a matched group who remained at legacy brokerages. Measure changes in listing volume, listing sides, average sale price, and luxury-market concentration before and after the move.

Suggested Method

- Treatment group: agents who switched to Compass in a given year. Control group: agents at the same prior brokerage, similar pre-period volume, who did not switch.
- Compare each group's change in volume/price metrics from the year before to the year after the treatment group's switch.
- State explicitly what this design can and cannot claim — it is suggestive of a Compass effect, not proof of causation, given the non-random nature of who switches firms.
- Apply IQR-based outlier flagging to ClosePrice before computing average-sale-price changes ($Q1/Q3 \pm 1.5 \times IQR$). Flag, don't delete — keep a full flagged dataset and a separate filtered analysis dataset — so a single \$50M estate sale doesn't distort your treatment-vs-control comparison.

Final Executive Memo

Write a 1–2 page memo, addressed to a brokerage CEO who has not seen your analysis, answering:

5. When did Compass begin materially gaining market share?
6. Which competitors lost the most share?
7. Did Compass recruit top-producing agents, and how do you know?
8. What share of growth came from acquisitions, agent migration, and organic growth?

9. Was Compass primarily a technology story or a talent-acquisition story — and what would change your mind?

The memo should lead with your answer, not your method. Save methodology detail for an appendix or the accompanying codebase.

Bonus Analysis — Backtest a Forecast

Using only data from your earliest available year through 2022, forecast whether Compass would surpass Keller Williams, Coldwell Banker, and RE/MAX in California listing volume. Then compare your forecast to actual 2023–2025 results.

- State your forecast method and assumptions before looking at 2023–2025 actuals — do not retrofit the model.
- Report forecast error and discuss what your model missed, if anything (e.g., a slowdown, an acquisition spree, a market downturn).

Evaluation Rubric

Criterion	Points	What strong work looks like
Data Engineering & Pipeline Quality	20	Filters, joins, and dedup logic are correct and reproducible from raw CRMLS extracts to the final analysis tables; queries are documented and re-runnable.
Data Provenance & QA Documentation	15	Clearly documents where the data came from (FTP pull vs. historical Trestle extraction), the date range actually covered, the agent/office ID-resolution method used, and every QA flag applied (date consistency, geographic, missing-value, outlier) before analysis begins.
Analytical Rigor	20	Migration matrices, concentration metrics, and the difference-in-differences design are methodologically sound, with assumptions and limitations stated explicitly.
Insight Quality	20	Findings directly answer the four Research Questions with evidence, not just description; surprising or counter-intuitive results are investigated rather than glossed over.
Visualization & Communication	15	Charts are correctly labeled, the right chart type is used for the message, and the Executive Memo is readable by a non-technical stakeholder in under five minutes.
Code Quality & Documentation	10	Code is organized, commented, and runs end-to-end without manual fixes; a README explains how to reproduce results.
Bonus: Forecast Accuracy	+10	Forecast trained only on data through 2022 is well-specified, and the comparison to 2023–2025 actuals includes a thoughtful discussion of forecast error.

Total: 100 points, plus up to 10 bonus points for the forecast backtest. 70+ is passing; 90+ is exemplary, presentation-ready work.

Suggested Timeline

Week	Focus	Milestone / Checkpoint
Week 1	Data ingestion, cleaning, and validation. Pull the CRMLS Sold extract, apply filters (Residential SFR), confirm the earliest usable year, and build your normalized agent/office composite keys.	Deliverable 1 draft: brokerage market share time-series chart.
Week 2	Build the agent-level panel. Rank top 500 agents annually; track brokerage affiliation changes; construct migration matrices.	Deliverable 2 complete; Deliverable 3 (top producer concentration) underway.
Week 3	Office-level growth analysis and acquisition detection logic; design and run the difference-in-differences comparison.	Deliverables 4, 5, and 6 complete with supporting visuals.
Week 4	Synthesize findings into the Executive Memo; run the bonus forecast (trained through 2022) and backtest against 2023–2025 actuals.	Final memo, full codebase, and 10-minute readout presentation submitted.

Submission Guidelines

- Submit a single repository or shared folder containing: source code/notebooks, the Executive Memo (PDF), and all charts as standalone image files.
- Include a README with setup instructions and a one-paragraph summary of your approach.
- Be prepared to present a 10-minute readout to your track lead, followed by Q&A — treat it as a real stakeholder briefing, not a status update.

Common Pitfalls

- Joining on agent or office name without a normalization step — name variants and collisions will silently corrupt your migration matrices. See “Agent & Office ID Resolution” above.
- Treating every disappearing small brokerage as a Compass acquisition — some simply closed or merged with a non-Compass entity.
- Reporting market share by transaction count when the question is about volume (\$), or vice versa, without being explicit about which metric is shown.
- Letting the difference-in-differences result overstate causality — always state the design's limitations alongside the finding.
- Assuming any specific start year (2015 or otherwise) is achievable before checking — confirm the earliest year CRMLS actually returns in Week 1, and state that confirmed start year explicitly in your methodology and memo.

Confidentiality

This case study uses real, live MLS transaction records, including agent and brokerage names. Treat the dataset as confidential business data: do not share, publish, or distribute it outside the internship program, consistent with IDX Exchange's standard data handling policy.

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